



Precision Additive Manufacturing & CNC Machining Solutions

PRECILAYER

Best-in-class precision machined and additive parts at scale, from India to the world. **No MOQs. No inventory.** Just speed, flexibility, and the future of making.

About Us

Changing the perception of Indian manufacturing

For too long, Indian manufacturing has been perceived as low-cost, low-quality. We're here to shatter that myth.

Precilayer represents the new generation of Indian manufacturing - one that competes on innovation, precision, and excellence

With a 20,000 square feet facility in Mumbai, India and a team of 18+ young engineers, we are onto this audacious mission

23+

Years Combined Experience

100+

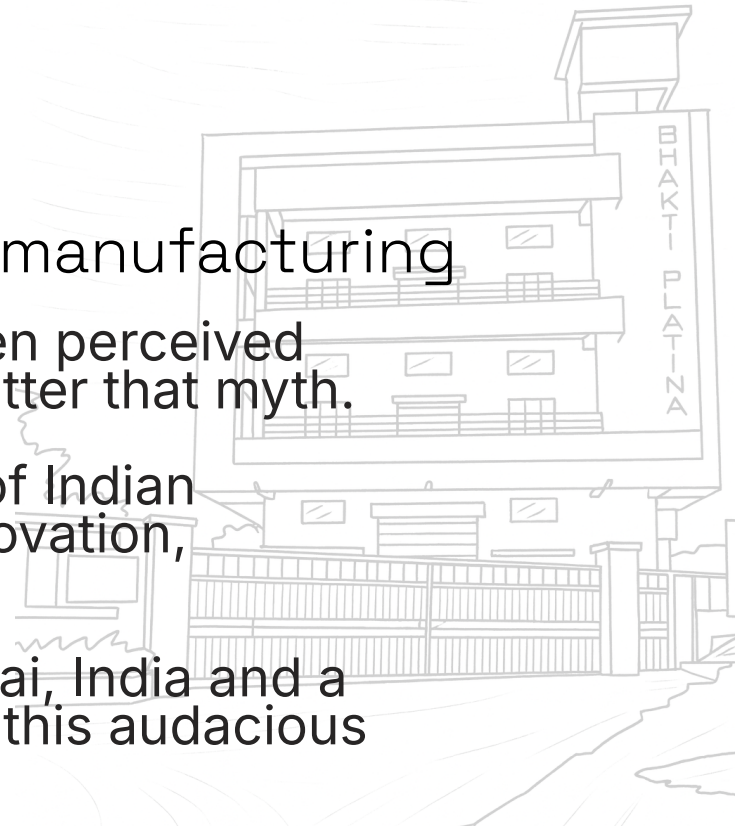
Global Clients Served

3

Continents Served

8+

Industry Verticals



The Founders



Roshan Kolhe
Co-Founder & Subtractive Manufacturing
Expert

Precision Engineering Excellence

When global giants need components that can't afford to fail, they rely on expertise like Roshan's. His track record spans critical applications where micron-level precision is non-negotiable.

Key clients: MedTech Leaders: Johnson & Johnson, Cook Medical, Merit Medical, Medtronic •

Semiconductor Giants: Intel, Analog Devices

Education: Masters from Dublin Institute of Technology • Bringing European precision standards to scale Indian manufacturing



Priyesh Mehta
Co-Founder & Additive Manufacturing design and
process Expert

Innovation Architecture

Where others see limitations, Priyesh sees possibilities. His collaboration portfolio spans three continents, transforming how automotive and aerospace leaders approach additive manufacturing.

Key clients: Drone Titans, Robotics startups in Switzerland, Oil and gas giants in UAE, Consumer startups in the US •

Client Helped: 350+ clients across the US, EU, Gulf regions

Education: Masters from The Pennsylvania State University • Published researcher • Turning 3D printing theory into manufacturing reality

Focus Industries: Precision Without Compromise



Space & Aerospace

Producing lightweight, high-strength components critical for satellite systems, propulsion, and avionics.



Drones & UAVs

Manufacturing intricate, durable parts that enhance performance, reduce weight, and ensure reliability for aerial applications.



Robotics & Automation

Creating precise gears, housings, and end effectors for industrial robots and automated systems, enabling smooth and accurate motion.



Mobility & Automotive

Delivering custom parts for electric vehicles, autonomous driving systems, and high-performance engines, focusing on efficiency and safety.



Healthcare & Medical Devices

Fabricating biocompatible and precision-engineered components for surgical instruments, implants, and diagnostic equipment.



Industrial & Manufacturing

Supplying robust and precise components for heavy machinery, specialized tooling, and custom industrial applications.

Polymer Additive Manufacturing Technologies



MultiJet Fusion/ Selective Laser Sintering



Stereolithography apparatus/Digital Light Projection

Parameter

Process Type

Primary Use

Key Advantages

Materials

**Maximum Part Size
(Single Shot)**

Large Part Capability

Post-Processing Options

Dimensional Tolerance

Typical Applications

MJF / SLS

Powder-based thermoplastic fusion

End-use production parts

High strength, functional, scalable, low cost per part at scale

Nylon PA12, Nylon PA12GB, TPU, PP, Nylon 11 CF, Carbon-fiber filled nylons

MJF: 380 × 284 × 380 mm
SLS: 700 × 380 × 580 mm

Split and weld possible

Dyeing, painting, insert fitment, vapor smoothing

±0.2 mm or ±0.2% (whichever is greater)

Enclosures, brackets, housings, structural parts

SLA / DLP

Vat photopolymerization

Prototyping, validation, mockups

High accuracy, surface finish, fine features

Standard, tough, durable, flexible, elastic, rigid, high-temperature, castable, ceramic-filled, biocompatible resins

800 × 800 × 600 mm

Split and weld possible

Ultrasonic welding, painting, insert fitment, metallic plating

±0.15 mm or ±0.15% (whichever is greater)

Thin-walled parts, medical models, jigs, fixtures, visual prototypes

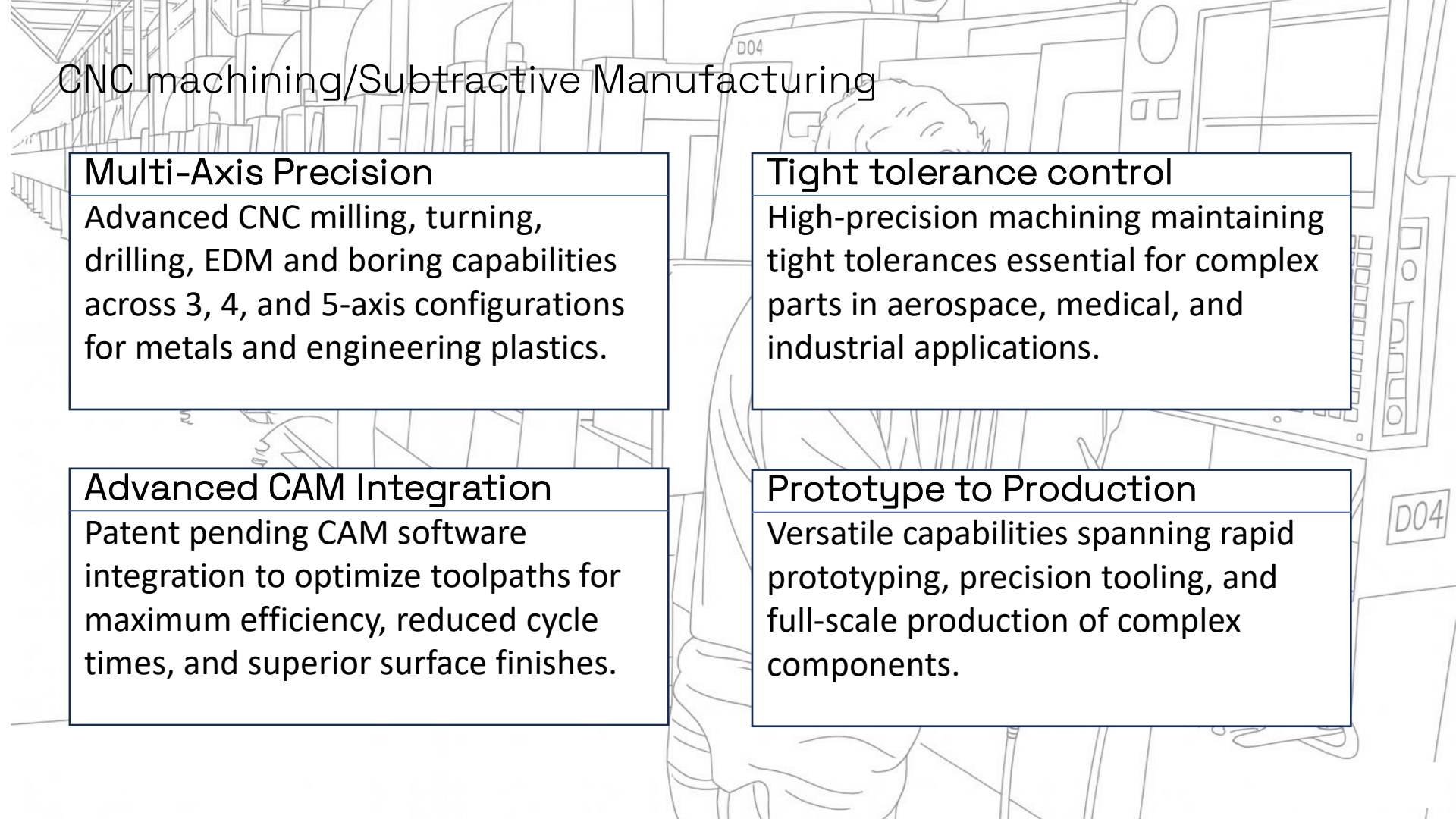


Metal Additive Manufacturing Technologies

 Direct Metal Laser Sintering  Electron Beam Melting

Parameter	DMLS (Direct Metal Laser Sintering)	EBM (Electron Beam Melting)
Process Type	Laser-based powder bed fusion	Electron beam powder bed fusion in vacuum
Primary Use	High-precision, complex end-use metal parts	High-performance, load-bearing metal parts
Key Advantages	High density, excellent dimensional stability, finest feature resolution	Fully dense parts, superior mechanical properties, ideal for reactive alloys
Materials	SS316L, SS 17-4PH, Maraging Steel, Inconel 718, 625, AlSi10Mg, Ti6Al4V Grade 5 & 23, Copper, custom alloys	Ti6Al4V Grade 5 & 23, CP Titanium (Grades 1-4), TiAl (Gamma Titanium Aluminide), Cobalt Chrome (CoCrMo), Inconel 718, 625, Copper (R&D), reactive alloys
Maximum Part Size (Single Shot)	400 × 400 × 450 mm	Typically, up to 350 × 350 × 380 mm (machine dependent)
Large Part Capability	Split and weld possible	Split and weld possible
Post-Processing Options	Heat treatment, anodizing, balancing for rotary parts, post-machining for GD&T	Heat treatment, HIP, machining, surface finishing
Dimensional Tolerance	±0.25 mm or ±0.25% (whichever is greater)	±0.3 mm or ±0.3% (typical, machine & material dependent)
Typical Applications	Aerospace brackets, housings, tooling, heat exchangers	Aerospace structural parts, medical implants, titanium load-bearing components





CNC machining/Subtractive Manufacturing

Multi-Axis Precision

Advanced CNC milling, turning, drilling, EDM and boring capabilities across 3, 4, and 5-axis configurations for metals and engineering plastics.

Tight tolerance control

High-precision machining maintaining tight tolerances essential for complex parts in aerospace, medical, and industrial applications.

Advanced CAM Integration

Patent pending CAM software integration to optimize toolpaths for maximum efficiency, reduced cycle times, and superior surface finishes.

Prototype to Production

Versatile capabilities spanning rapid prototyping, precision tooling, and full-scale production of complex components.

CNC machining/Subtractive Manufacturing

We work with an extensive selection of engineering-grade materials, carefully chosen to suit the unique requirements of various industries. Our deep knowledge allows us to recommend and process the ideal material for your project, balancing mechanical properties, environmental resistance, and cost-effectiveness.

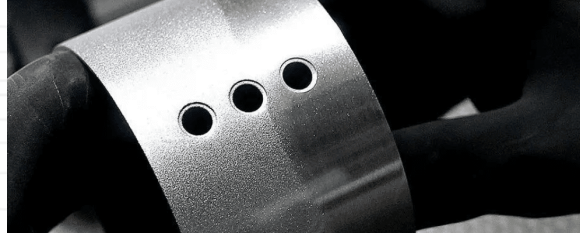
Material Category	Key Properties	Common Grades / Types	Typical Applications
Aluminum Alloys	Lightweight, corrosion-resistant, high thermal conductivity	5000, 6000, 7000 series, specialty aerospace alloys	Frames, housings, heat sinks, lightweight structures
Stainless Steel & Tool Steels	High strength, toughness, wear & corrosion resistance	Carbon steels (C45, A36), Alloy steels (4140, 4340), Tool steels (D2, H13), SS 300 & 400 series	Structural parts, shafts, tooling, corrosion-resistant components
Hard & Super Alloys	High temperature strength, oxidation resistance, extreme durability	Titanium, Inconel, Nickel-based superalloys, Cobalt alloys	Aerospace, power generation, medical, high-temperature components
Brass & Copper Alloys	Excellent machinability, electrical & thermal conductivity	C36000 Free-Cutting Brass, C38500 Architectural Brass, C46400 Naval Brass, C23000 Red Brass, C11000 ETP Cu, C14500 Tellurium Cu, C17200 Beryllium Cu	Electrical components, marine parts, fittings, precision hardware
Carbides & Ceramics	Extreme hardness, wear & abrasion resistance	Tungsten Carbide, Titanium Carbide, Tantalum Carbide, Ceramic composites	Cutting tools, wear inserts, high-abrasion environments
Engineering Plastics & Polymers	Lightweight, corrosion-resistant, machinable, low friction	PEEK, PTFE, Nylon, ABS, Delrin, Polycarbonate, Acetal, UHMW	Insulators, low-friction parts, housings
Other / Specialty Materials	Custom performance, tailored properties	Composites, exotic alloys, custom blends	R&D parts, high-performance assemblies

Finishes available

Metal Grinding



Media Blasting



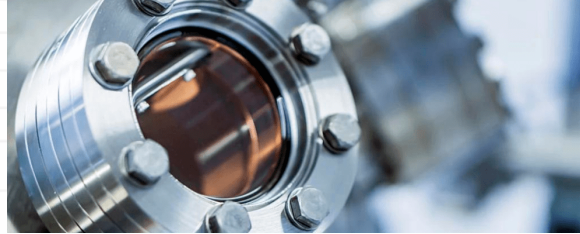
Aluminum Anodizing



Metal Passivation



Electropolishing



Metal Polishing



Metal Tumbling



Powder Coating



Tufram & Nedox (SF2)



Quality Assurance and Processes

At Precilayer, quality is not just a standard; it's an ingrained philosophy that permeates every stage of our manufacturing process. We implement stringent quality control measures and robust process management for both our additive manufacturing and CNC machining operations, ensuring every component meets the highest industry specifications and customer expectations.

01

Material Validation & Preparation

All incoming raw materials undergo meticulous inspection and verification against certifications (e.g., Material Test Reports). For additive manufacturing, this includes powder analysis, while for CNC, it ensures material integrity and correct alloy composition before any machining begins.

03

Post-Processing & Intermediate Inspection

After initial manufacturing, components often undergo various post-processing steps such as heat treatment, stress relief, or surface finishing. Each of these stages is followed by dedicated inspection points, including visual checks, hardness testing, and dimensional measurements to ensure process integrity.

Comprehensive Documentation & Traceability

We provide full documentation for every project, including Certificates of Conformance (CoC), detailed inspection reports, material traceability records, and custom quality plans. Our adherence to industry standards like ISO 9001 and AS9100 ensures complete transparency and accountability throughout the manufacturing lifecycle.

02

In-Process Monitoring & Control

Our advanced machines are equipped with real-time monitoring systems. For additive processes, this involves in-situ optical pyrometry and layer-by-layer inspection. In CNC, it includes constant tool wear compensation, automated probe measurements, and statistical process control (SPC) to maintain dimensional accuracy and surface finish during production.

04

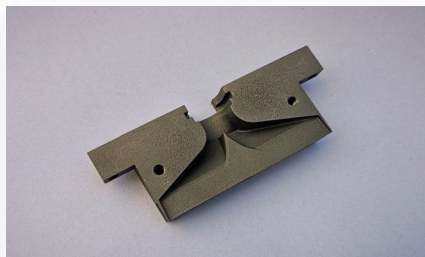
Final Metrology & Non-Destructive Testing (NDT)

Every finished part undergoes comprehensive final inspection using state-of-the-art metrology equipment, including Coordinate Measuring Machines (CMM), optical scanners, and profilometers. We also employ Non-Destructive Testing (NDT) methods like X-ray computed tomography (CT scanning) for internal defect detection in additive parts, or dye penetrant inspection for surface flaws.

Proven Performance : Additive Manufacturing

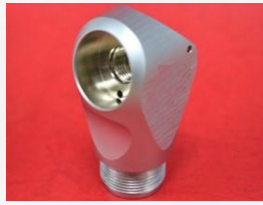


Industry Medical
Material: 17-4



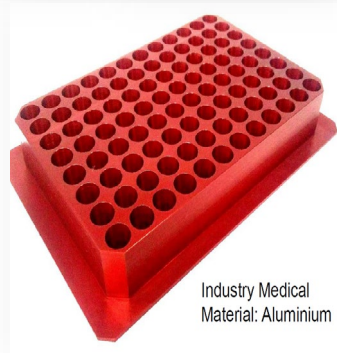
Aerospace Lightweighting

Utilized DMLS for a topologically optimized titanium bracket, achieving a **40% weight reduction** for critical aircraft components while enhancing structural integrity. This project showcased our ability to push boundaries in complex geometry and material performance. Precilayer has secured a 3-year contract with this ESA certified space company since we helped the design team achieve significant savings, both in costs and weight



Medical Device Innovation

Engineered a patient-specific orthopedic implant using Electron Beam Melting, featuring intricate internal porosity for improved bone integration. Our precision ensured **biocompatibility and exact fit**, significantly aiding post-operative recovery.



Industry Medical
Material: Aluminium

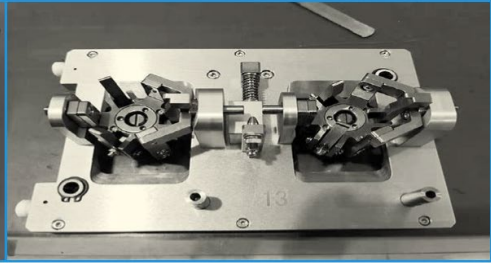
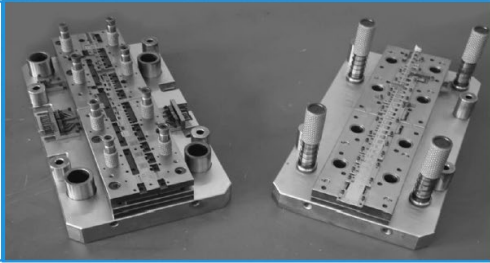
Drone Turnkey solution

Precilayer is selected for a 18-month contract to manufacture mechanical components for a Zurich based drone company and assemble the entire drone with sourcing and electronics support provided by a Koren company. This helped the Supply chain team reduce **62% of costs** and have a standard lead time of 1 week for their drone deliveries.

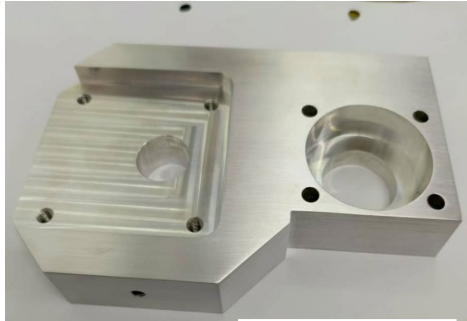
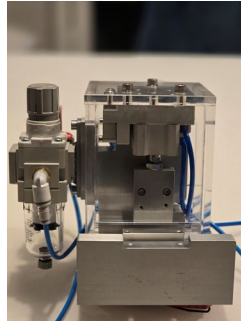
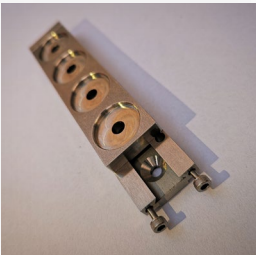
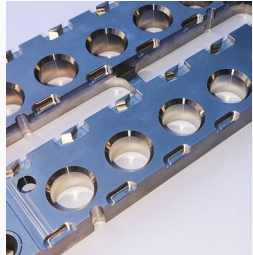
Proven Performance : CNC machining

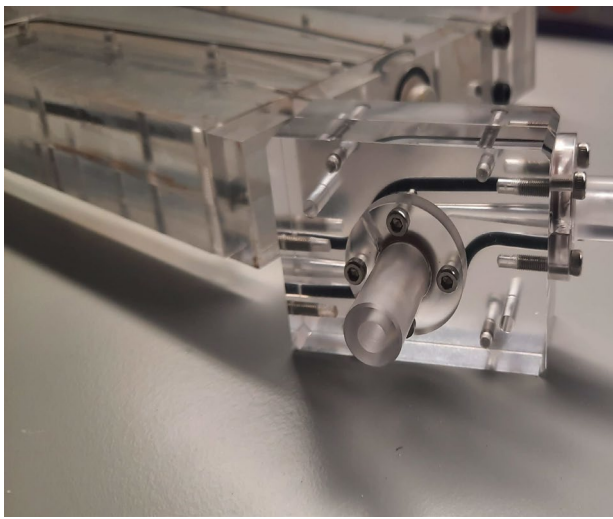


Small Assemblies



Fabrication



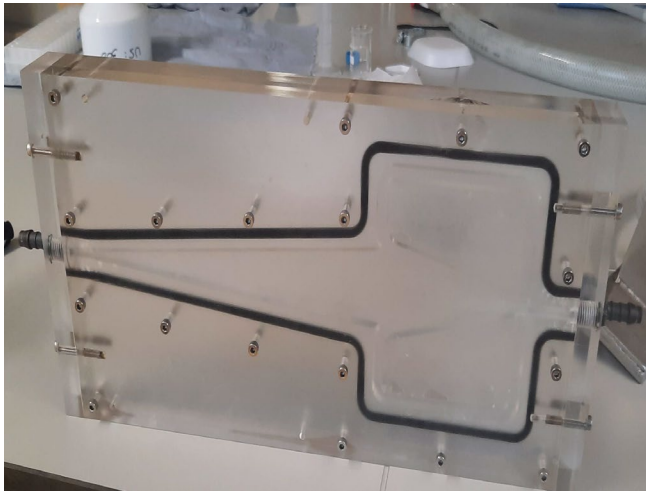


Automotive tooling customer

Precilayer helped an automotive tooling customer relocate their supply chain from China to India – the shift was strategic for the company and helped them save **28%** on import tariffs

Precision robotics components

Machined **high-tolerance** aluminum engine components on 5-axis CNC machines, meeting strict dimensional requirements of $\pm 0.005\text{mm}$. This ensured optimal performance and extended lifespan for their vehicles.



Complex Mold for consumer electronics

Developed an intricate mold cavity from hardened tool steel using wire and sinker EDM, achieving mirror-like surface finishes and sub-micron feature accuracy crucial for advanced consumer electronic housings. Helped them develop a secondary **source outside of China**

Industrial Valve Systems

Manufactured high-pressure stainless steel valve bodies with CNC turning, maintaining **critical sealing surfaces** and internal passages. Our components ensured reliability and safety in demanding industrial fluid control applications.

Advantages of Partnering with Precilayer



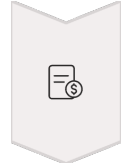
Rapid Scaling

Seamless transition from rapid prototyping to production scale with flexible batch sizes meeting your timeline requirements. Our quote turnaround time is 24 hours for projects < 10 parts and 48 hours for projects > 10 parts



Design Freedom

Unprecedented design flexibility enabling complex, lightweight, and integrated parts impossible with traditional manufacturing. We understand the need for engineering changes mid project and can accommodate those swiftly.



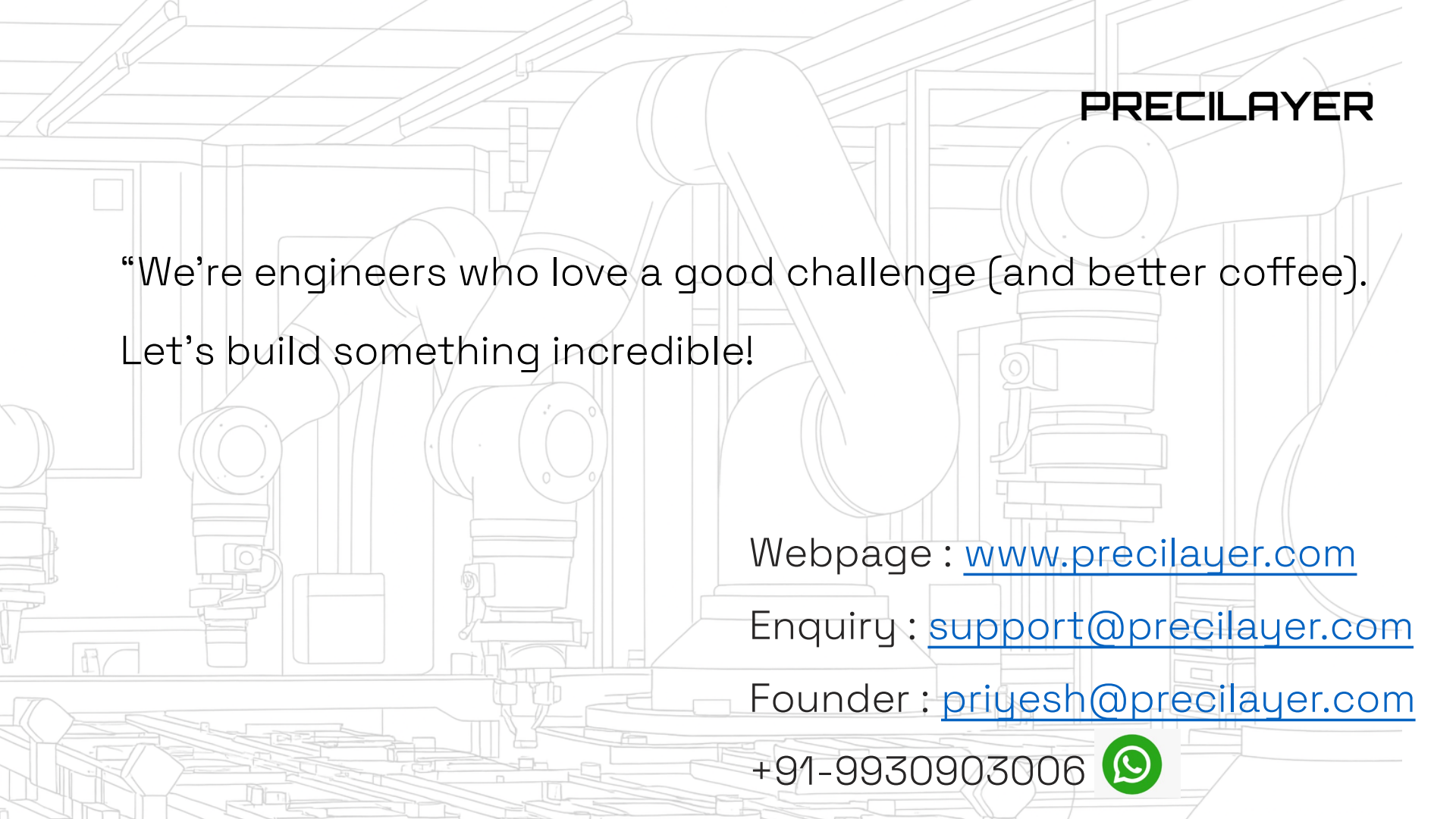
Cost Efficiency

With the attractive cost of resources like land, manpower and electricity in India, we provide a strong cost-effective option for western companies seeking to expand their manufacturing sourcing in India. The geopolitical advantage offered by India with government incentives and free trade agreements also add to the savings



Skilled Manpower

Every year, India adds 1.5 million STEM engineers to its workforce. At Precilayer, we handpick and nurture those who are truly passionate about building things and getting hands-on on the shop floor.



PRECILAYER

“We’re engineers who love a good challenge (and better coffee).
Let’s build something incredible!

Webpage : www.precilayer.com

Enquiry : support@precilayer.com

Founder : priyesh@precilayer.com

+91-9930903006

